

GENERAL AUDIENCE ABSTRACT

Multi-year analysis is often used to understand how characteristics generally relate to each other. In meteorology, general characteristic studies can, for example, be applied to how supercell size relates to tornado intensity or how stratiform reflectivity relates to rainfall rates. This study was designed to see if it would be possible to observe lightning and storm cell characteristics over a short time period and later expand to larger datasets for a more thorough understanding. Analyzing lightning characteristics within a multi-day event has been done for severe events across the world, but the ability to analyze years of data can provide a general idea of how storms impact lightning over a specific region. This project made use of Adapt, an orchestrated framework for total analysis of a storm cell, alongside xlma-python, a program for viewing lightning data from a lightning mapping array network, to see if a multi-year analysis was possible. After analyzing one event from May 2nd-3rd, 2025 over north Alabama, patterns were observed relating increases in lightning size and strength to differential reflectivity values between 2-4 db as well as an increasing trend in lightning activity following storm identification, even when the storms themselves do not appear to strengthen. The results of this study provide little insight into how storm processes modify lightning characteristics, but it proves that this kind of analysis is possible and can be scaled up over multiple years and regions. With a larger dataset, there will be more storms with unique characteristics that can be observed to see how those underlying characteristics influence storm and lightning development. Further developing Adapt into a thorough analysis tool allowing for association of measurements by different instruments with storm cells will continue to make larger analysis worthwhile and improve forecasting and early warning techniques through larger studies.